=== UNIT 3: CELLULAR BIOENERGETICS, MITOCHONDRIA & CHLOROPLAST ===

>> 3.1 Mitochondrial Ultrastructure and Chemiosmotic Coupling

Mitochondria are the primary bioenergetic hubs of eukaryotic cells, generating >90% of cellular ATP through oxidative phosphorylation.
* Double Membrane Architecture: Outer mitochondrial membrane (OMM) contains large pore-forming beta-barrel proteins called VDAC (porins), rendering it freely permeable to ions and solutes <5 kDa. The inner mitochondrial membrane (IMM) is impermeable to small ions and protons, exhibiting a 75% protein-by-weight ratio and enriched in cardiolipin (diphosphatidylglycerol) containing 4 acyl chains. IMM forms extensive inward invaginations termed cristae, organized by the MICOS complex and ATP synthase dimers.
* Peter Mitchell's Chemiosmotic Hypothesis (Nobel Prize 1978): As electrons pass down the respiratory chain, exergonic redox reactions are coupled to the active extrusion of protons (H+) from the mitochondrial matrix into the intermembrane space (IMS). This establishes a proton motive force (pmf or delta p): delta p = delta psi - (2.303 * R * T / F) * delta pH. At 37 deg C, delta p ~ -180 to -220 mV, comprising a major electrical membrane potential (delta psi ~ -160 mV, matrix negative) and a minor pH gradient (delta pH ~ 0.75-1.0 unit, matrix alkaline).

>> 3.2 The Electron Transport Chain (Complexes I - IV)

* Complex I (NADH:Ubiquinone Oxidoreductase): Massive 1 MDa L-shaped enzyme (45 subunits). Accepts 2 electrons from NADH via FMN cofactor, transfers them along 8-9 Fe-S clusters to reduce ubiquinone (Q) to ubiquinol (QH2). Translocates 4 H+ across the IMM per NADH oxidized via long horizontal coordinating alpha-helix.
* Complex II (Succinate Dehydrogenase): The only membrane-bound component of the citric acid cycle. Oxidizes succinate to fumarate via FAD and 3 Fe-S clusters, reducing Q to QH2 without pumping any protons.
* Complex III (Cytochrome bc1 Complex / Ubiquinol:Ferricytochrome c Oxidoreductase): Operates via the Q-cycle mechanism. For every 2 ubiquinol molecules oxidized, 4 protons are extruded into the IMS, 2 cytochrome c molecules are reduced, and 1 ubiquinone is regenerated. Contains Cytochrome b (bL and bH hemes), Rieske [2Fe-2S] center, and Cytochrome c1.
* Complex IV (Cytochrome c Oxidase): Terminal respiratory oxidase (13 subunits, contains heme a, heme a3, CuA, and CuB binuclear centers). Transfers 4 electrons from 4 reduced cytochrome c molecules to molecular oxygen: O2 + 4 H+ (matrix) + 4 e- -> 2 H2O. Pumps an additional 4 H+ into the IMS per O2 reduced.
* Stoichiometry: Total protons pumped per pair of electrons: NADH -> O2 pumps 10 H+ (4 by CI, 4 by CIII, 2 by CIV); FADH2 -> O2 pumps 6 H+ (4 by CIII, 2 by CIV).

>> 3.3 ATP Synthase (Complex V) Rotary Catalysis

* Molecular Architecture: Composed of two functional domains: membrane-embedded F0 (proton channel formed by oligomeric c-ring and subunit a) and soluble matrix F1 (catalytic headpiece with alpha3-beta3 hexamer surrounding the asymmetric gamma-epsilon central stalk, stabilized by peripheral b-d-F6-OSCP stator).
* Paul Boyer's Binding Mechanism: The 3 catalytic beta subunits adopt 3 alternating conformations: Open (O, very low affinity for nucleotides), Loose (L, binds ADP + Pi loosely), and Tight (T, binds substrates tightly and catalyzes spontaneous ATP synthesis). Proton flow through subunit a drives rotation of the c-ring and attached central gamma-stalk. As gamma rotates 360 degrees, it forces sequential conformational transitions (O -> L -> T -> O), synthesizing 3 ATP molecules per complete revolution.

>> 3.4 Chloroplast Ultrastructure and Photophosphorylation

* Thylakoid Membrane Organization: Grana thylakoids (stacked disks enriched in Photosystem II) and stroma thylakoids (unstacked membranes enriched in Photosystem I and ATP synthase). Light-harvesting complexes (LHC-II) capture photons and funnel resonance energy to reaction center P680.
* Photosystem II & Oxygen-Evolving Complex: P680* excitation releases electrons to pheophytin, plastoquinone A, and plastoquinone B. P680+ is the most powerful biological oxidizing agent (E0' ~ +1.25 V), capable of water photolysis catalyzed by the Mn4CaO5 cubane cluster: 2 H2O -> O2 + 4 H+ + 4 e-.
* Z-Scheme Electron Flow: Electrons move from PS II -> Plastoquinone (PQ) -> Cytochrome b6f complex (pumps H+ into thylakoid lumen) -> Plastocyanin (PC) -> Photosystem I (P700) -> Ferredoxin (Fd) -> Ferredoxin-NADP+ reductase (FNR) to generate NADPH.
* Calvin-Benson Cycle (Dark Reactions): Carbon fixation in the stroma by RuBisCO (Ribulose-1,5-bisphosphate carboxylase/oxygenase): RuBP (5C) + CO2 -> two 3-phosphoglycerate (3C). Followed by reduction to glyceraldehyde-3-phosphate (G3P) consuming 9 ATP and 6 NADPH per 3-carbon sugar exported.

[EXAMPLE 1: P/O Ratio and Theoretical ATP Yield per Glucose Molecule]

Problem Context: Using modern chemiosmotic stoichiometries (c-ring subunit count = 8 in mammalian mitochondria, 1 H+ consumed per ATP by the adenine nucleotide translocase/phosphate translocase symporter), calculate the precise P/O ratio for NADH and FADH2, and deduce the net ATP yield from the complete aerobic oxidation of 1 molecule of D-glucose.
Proton Accounting:
- 1 complete revolution of ATP synthase c-ring yields 3 ATP molecules.
- For a mammalian c-ring with 8 c-subunits, 8 H+ traverse F0 per revolution.

- Phosphate import requires symport of 1 H+ with H2PO4- per ATP synthesized.
- Total H+ required per ATP produced = (8 H+ / 3 ATP) + 1 H+ = 2.67 + 1.00 = 3.67 H+ per ATP.

P/O Ratio Calculation:

- NADH oxidation pumps 10 H+: P/O ratio = 10 / 3.67 = 2.72 (conventionally taken as 2.5 ATP/NADH).
- FADH2 oxidation pumps 6 H+: P/O ratio = 6 / 3.67 = 1.63 (conventionally taken as 1.5 ATP/FADH2).

Total Net ATP per Glucose:

1. Glycolysis: 2 ATP (substrate-level) + 2 NADH (cytosolic). Via malate-aspartate shuttle (2 * 2.5 = 5 ATP).
2. Pyruvate Dehydrogenase: 2 NADH (mitochondrial matrix) = 2 * 2.5 = 5 ATP.
3. Krebs Cycle (2 turns): 2 GTP/ATP (substrate-level) + 6 NADH (6 * 2.5 = 15 ATP) + 2 FADH2 (2 * 1.5 = 3 ATP) = 20 ATP.

Net Total Yield = 2 + 5 + 5 + 20 = 32 ATP per glucose (or 30 ATP if glycerol-3-phosphate shuttle is used).

Conclusion: Chemiosmotic coupling yields 30-32 ATP per glucose, substantially lower than the older theoretical estimate of 36-38 ATP which ignored proton costs of solute translocases.

[EXAMPLE 2: Chemical Uncoupling Agents and Brown Adipose Thermogenesis]

Problem Context: 2,4-Dinitrophenol (DNP) is a toxic lipophilic weak acid that acts as a protonophore uncoupler, whereas Brown Adipose Tissue (BAT) uses physiological uncoupling via Thermogenin (UCP1). Contrast their mechanisms, effects on oxygen consumption, pmf, and body temperature.

Molecular Mechanism of DNP: DNP has a pKa of 7.2. In the acidic intermembrane space (pH ~7.0), it accepts a proton to become electrically neutral, readily diffuses through the hydrophobic IMM lipid bilayer, and dissociates its proton in the alkaline matrix (pH ~8.0).

Consequences of DNP Administration:

- Dissipation of Proton Motive Force: delta p collapses to near zero without inhibiting electron transport.
- Unrestrained Respiration: In the absence of back-pressure from the proton gradient, respiratory Complexes I-IV operate at maximal velocity. Oxygen consumption (VO2) skyrockets.
- ATP Depletion and Hyperthermia: Complex V cannot synthesize ATP because the proton gradient is gone. The entire redox free energy of substrates is released purely as thermal energy (heat). Clinically causes fatal hyperthermia (>42 deg C), metabolic acidosis, and diaphoresis.

Physiological Thermogenin (UCP1) in BAT: UCP1 is a regulated transmembrane channel activated by free fatty acids (following beta-3 adrenergic stimulation by norepinephrine in infants and cold-adapted mammals). It channels protons across the IMM, converting electrochemical potential into non-shivering thermogenesis to maintain infant core body

temperature safely.

[EXAMPLE 3: Site-Specific Inhibitors of the Respiratory Chain]

Problem Context: Analyze the biochemical diagnostic response (redox state of cytochromes and oxygen consumption rate) when the following four distinct metabolic poisons are added sequentially to an in vitro suspension of respiring liver mitochondria fed with malate and glutamate: (A) Rotenone, (B) Antimycin A, (C) Cyanide (CN-), and (D) Oligomycin.

Analysis of Inhibitors:

1. Rotenone (Plant isoflavone / Piscicide): Binds Fe-S cluster PSST/ND1 in Complex I, preventing electron transfer to Ubiquinone. NADH remains fully reduced; ubiquinone and cytochromes b, c1, c, a/a3 become oxidized. VO2 drops to zero. Can be bypassed by adding succinate (feeds Complex II directly).
2. Antimycin A (Streptomyces antibiotic): Binds Qi site of Complex III, halting the Q-cycle. Cytochromes b remain reduced; cytochromes c1, c, and a/a3 become oxidized. VO2 ceases.
3. Cyanide (KCN / CO / Azide): Binds with ultra-high affinity to the Fe3+ of heme a3 in Complex IV, blocking O2 reduction. All upstream electron carriers (Complexes I, II, III, cytochromes) become 100% reduced. VO2 drops instantly to zero. Causes immediate cellular suffocation.
4. Oligomycin (Streptomyces macrolide): Binds directly to the c-subunit ring of F0 in ATP synthase, physically blocking the proton channel. Protons cannot re-enter the matrix; delta p rises to a maximum until the proton back-pressure halts further electron pumping by Complexes I-IV. Respiration stops, but can be restored immediately by adding DNP uncoupler.

Conclusion: Chemical inhibitors confirm the linear sequential nature of the electron transport chain and demonstrate mandatory tight coupling between ATP synthase and redox proton pumping.

[EXAMPLE 4: Cyclic vs. Non-Cyclic Photophosphorylation Energetics]

Problem Context: In higher plant chloroplasts, evaluate why non-cyclic photophosphorylation alone is insufficient to satisfy the stoichiometric ATP:NADPH demand of the Calvin cycle, and explain how cyclic photophosphorylation resolves this bioenergetic shortfall.

Calvin Cycle Demand: Fixation of 3 CO_2 molecules to produce 1 net G3P triose phosphate requires exactly 6 NADPH and 9 ATP (an ATP:NADPH ratio of 1.50).

Non-Cyclic Electron Transport (Z-Scheme) Output:

- 2 photons absorbed per electron (1 by PS II, 1 by PS I) = 4 photons per electron pair.
- Transport of 4 electrons from 2 H_2O to 2 $NADP^+$ pumps 12 H^+ into the thylakoid

lumen (4 H+ from water photolysis + 8 H+ via Q-cycle at Cytochrome b6f).
- Chloroplast ATP synthase c-ring has 14 subunits, requiring 14 H+ per 3 ATP (4.67 H+ per ATP).
- 12 H+ / 4.67 H+ per ATP = 2.57 ATP synthesized per 2 NADPH produced.
- Yielded ATP:NADPH ratio = 2.57 / 2 = 1.28 (significantly below the required 1.50).
Cyclic Electron Flow Resolution:
- When the chloroplast ATP/ADP ratio falls, reduced ferredoxin (Fd) diverts electrons away from FNR back to the Cytochrome b6f complex (via PGR5/PGRL1 or NDH complex).
- Electrons cycle: Fd -> PQ -> Cyt b6f -> PC -> P700 (PS I) -> Fd.
- Net result: Protons are continuously pumped into the thylakoid lumen generating a proton gradient and driving ATP synthesis without any photolysis of water and without generating NADPH.
Conclusion: By dynamically adjusting the ratio of cyclic to non-cyclic electron flow, the photosynthetic apparatus precisely matches the ATP:NADPH requirements of the dark reactions.

>> 3.5 University Review & Examination Questions

* Q1: Diagram and describe the binding change mechanism of F0F1-ATP synthase as formulated by Paul Boyer.
Answer: The catalytic F1 domain comprises an alpha3-beta3 hexamer. The three catalytic beta subunits exist in three different conformations: Open (O, very low substrate affinity; newly formed ATP dissociates), Loose (L, binds ADP + Pi with moderate affinity but cannot catalyze synthesis), and Tight (T, binds substrates tightly and drives spontaneous condensation of ADP and Pi into ATP without external energy input). The asymmetric central gamma stalk is physically attached to the rotating membrane-embedded c-ring of F0. Driven by proton flow down the electrochemical gradient through subunit a, the gamma stalk rotates in 120-degree increments. Each 120-degree rotation forces a cooperative conformational transition: T -> O (ATP released), L -> T (ATP synthesized), and O -> L (new ADP+Pi bound). A complete 360-degree rotation synthesizes and releases 3 ATP molecules.
* Q2: Explain the structural and thermodynamic basis of the Q-cycle in the Cytochrome bc1 complex (Complex III).
Answer: The Q-cycle allows a two-electron carrier (ubiquinol, QH2) to transfer electrons one at a time to a one-electron carrier (cytochrome c) while maximizing proton translocation. In the first half-cycle, QH2 binds the Qo site near the intermembrane space and releases 2 H+ into the IMS. Its first electron reduces the Rieske [2Fe-2S] cluster and cytochrome c1 to reduce cytochrome c. Its second electron passes via low-potential heme bL and high-potential heme bH to reduce a bound ubiquinone at the Qi site into a semiquinone radical (Q.-). In the second half-cycle, a second QH2 is oxidized at Qo, releasing 2 more H+ into IMS; one electron reduces a second cytochrome c, while the other reduces the semiquinone radical at Qi, consuming 2 matrix H+ to regenerate QH2. Net reaction: 1 QH2 oxidized, 2 cyt c reduced, 4 H+ pumped to IMS.

>> 3.6 Key Takeaways and Revision Points

* Peter Mitchell’s chemiosmotic hypothesis: electron transport drives proton extrusion, creating a proton motive force (pmf = delta psi - 59 delta pH).
* Complex I pumps 4 H+, Complex III pumps 4 H+, Complex IV pumps 2 H+ per electron pair from NADH (total 10 H+).
* Modern ATP yield is 2.5 ATP/NADH and 1.5 ATP/FADH2, giving 30-32 ATP per glucose.
* Rotenone, Antimycin A, Cyanide, and Oligomycin sequentially dissect respiratory chain coupling.